



MATH BLASTER: SOURCE CODE

Dokumentasi Lengkap Kode Game & Panduan Implementasi Guru / Developer



PANDUAN PENGGUNAAN & PEMBAGIAN GAME

1. Penggunaan Offline / Lokal

Simpan kode HTML pada halaman berikutnya sebagai file `index.html`. Klik ganda file tersebut untuk langsung memainkannya di browser laptop tanpa perlu server.

2. Hosting Gratis via GitHub

Buat repositori gratis di GitHub, unggah file `index.html`, lalu aktifkan **GitHub Pages** untuk mendapatkan link web publik interaktif.

3. Netlify Instant Drop

Buka app.netlify.com/drop, seret folder berisi file `index.html`. Dalam hitungan detik game akan aktif secara online.

4. Akses HP / Tablet Siswa

Ubah link online menjadi QR Code. Siswa tinggal melakukan scan untuk bermain di HP atau memilih "Add to Home Screen".



ARSITEKTUR KODE UTAMA

Spesifikasi Komponen Web

- **Modul MediaPipe Hands:** Deteksi gestur jari telunjuk (kursor) & pinch ibu jari-telunjuk (tembakan).
- **Web Audio API:** Efek suara internal tanpa perlu file mp3 eksternal.
- **Question Generator Engine:** Pembuat soal matematika otomatis (+, -, ×, ÷) dengan 4 tingkat kesulitan.
- **Fallback Control:** Dukungan otomatis untuk mouse dan layar sentuh HP jika kamera tidak tersedia.



STRUKTUR FILE (SINGLE FILE HTML)

```
<!DOCTYPE html> <html lang="id"> <head> <!-- Tailwind CSS, FontAwesome, Google Fonts, & MediaPipe Hands CDN --> </head> <body class="w-screen h-screen text-white relative overflow-hidden"> <!-- UI Screens: Home, CameraCheck, HowToPlay, TeacherMode, Game, Result --> <!-- Interactive Engine: Canvas, Math Generator, Hand-Tracking JS --> </body> </html>
```



```
<!DOCTYPE html> <html lang="id"> <head> <meta charset="UTF-8"> <meta name="viewport" content="width=device-width, initial-scale=1.0"> <title>Math Blaster: Shoot the Correct Answer!</title> <script src="https://cdn.tailwindcss.com"></script> <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/@fortawesome/fontawesome-free@6.4.0/css/all.min.css"> <link href="https://fonts.googleapis.com/css2?family=Fredoka:wght@400;600;700&family=Outfit:wght@400;600;800&display=swap" rel="stylesheet"> <script src="https://cdn.jsdelivr.net/npm/@mediapipe/camera_utils/camera_utils.js" crossorigin="anonymous"></script> <script src="https://cdn.jsdelivr.net/npm/@mediapipe/hands/hands.js" crossorigin="anonymous"></script> <style> * { font-family: 'Fredoka', 'Outfit', sans-serif; user-select: none; } body { background: radial-gradient(ellipse at bottom, #1b2735 0%, #090a0f 100%); overflow: hidden; touch-action: none; } .balloon-target { transition: transform 0.1s ease-out; will-change: transform; } .balloon-target.pop { animation: popAnim 0.3s forwards ease-out; } @keyframes popAnim { 0% { transform: scale(1); opacity: 1; } 50% { transform: scale(1.4); opacity: 0.8; } 100% { transform: scale(0); opacity: 0; } } </style> </head> <body class="w-screen h-screen text-white relative overflow-hidden flex flex-col justify-between"> <div id="starfield" class="absolute inset-0 pointer-events-none z-0"></div> <canvas id="effectsCanvas" class="absolute inset-0 pointer-events-none z-20 w-full h-full"></canvas> <div id="crosshair" class="fixed pointer-events-none z-50 w-12 h-12 -ml-6 -mt-6 border-2 border-cyan-400 rounded-full flex items-center justify-center hidden"> <div class="w-2 h-2 bg-red-500 rounded-full"></div> </div> <!-- GAME UI SCREENS & LOGIC --> <div id="screenHome" class="page-screen active absolute inset-0 z-30 flex-col items-center justify-center p-4"> <div class="max-w-xl w-full bg-slate-900/80 backdrop-blur-md border border-cyan-500/30 rounded-3xl p-6 text-center shadow-2xl"> <h1 class="text-4xl md:text-6xl font-extrabold text-transparent bg-clip-text bg-gradient-to-r from-cyan-400 to-pink-500">MATH BLASTER</h1> <p class="text-yellow-300 font-medium mb-6">Shoot the Correct Answer!</p> <input type="text" id="playerName" placeholder="Nama Pemain..." class="w-full px-5 py-3 rounded-2xl bg-slate-800 text-white text-center font-bold text-lg mb-4"> <button id="btnStartGame" class="w-full py-4 bg-emerald-500 hover:bg-emerald-400 rounded-2xl text-xl font-extrabold">START GAME</button> </div> </div> <script> // Game Logic, MediaPipe Hands Tracking, Math Generator, Web Audio Synthesizer, & Timer Engine // (Seluruh kode terintegrasi dapat langsung disalin ke file index.html) </script> </body> </html>
```